NOURHAN ATEF

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SUMMARY

Computer Science and AI undergraduate at Cairo University and Samsung Innovation Campus AI Track graduate, focused on machine learning in Python. Hands-on experience training and comparing deep learning models (Keras/TensorFlow), time-series forecasting (ARIMA, RNN/LSTM), recommender systems and churn prediction, and building LLM pipelines (Hugging Face, local LLMs via Ollama, RAG, agentic workflows). Comfortable with experiment tracking (MLflow), clear evaluation write-ups, and Git. Seeking to contribute to ML model training, evaluation, and agentic tooling in a research-driven team.

EDUCATION

B.Sc. Computer Science & Artificial Intelligence | Cairo University, FCAI 2025 – Present

- Major: Data Science
- Relevant Coursework: Python Programming, Discrete Mathematics, Data Structures and Algorithms, structured C++ and OOP in C++, Operation Research and Decision Support, Electronics & Embedded Systems, Logic Design, Calculus and ODEs

EXPERIENCE

AI Track Intern (Graduate) | Samsung Innovation Campus (SIC), Life Makers Foundation July 2026 – November 2026

- Completed an intensive AI program covering probability and statistics, linear algebra, data preprocessing (NumPy, Pandas), machine learning, deep learning, NLP, LLMs, and AI automation
- Machine learning: linear and logistic regression (including from scratch with gradient descent), regularization (Ridge), KNN, SVM, Naive Bayes, decision trees and ensembles (Random Forest), model evaluation and selection, time series (ARIMA), clustering (K-Means, DBSCAN), PCA
- Deep learning: TensorFlow and Keras, CNNs for image classification, RNN/LSTM for sequence forecasting, image processing
- LLMs and agents: NLP preprocessing and sentiment analysis, open-source LLMs, fine-tuning, Retrieval-Augmented Generation (RAG), prompt engineering, agentic AI, workflow automation (Make, n8n)
- MLOps: experiment tracking with MLflow, model deployment with Flask, Streamlit, and Gradio
- Delivered three capstone projects and a team graduation project, presented to instructors

PROJECTS

H&M Personalized Fashion Recommendations – "Beyond The Purchase" | Python | Recommender Systems | Churn Prediction | SIC Graduation Project, Team 9 | 2026

- Built a two-stage recommender system and a customer churn prediction model on the H&M Kaggle dataset (large-scale transaction data, sampled to 10% of customers for tractable experimentation)
- Cleaned and joined customer, article, and transaction tables; led the exploratory analysis, turning the data into 10 business questions with supporting visualizations
- Evaluated model performance and presented methodology, results, and business impact in a team deck; collaborated through a shared Kaggle notebook

Hybrid Sentiment Inference Pipeline (OmniPulse AI) | Python | Hugging Face | Ollama | LLMs | 2026

- Built an inference pipeline on 50K IMDB reviews comparing a Hugging Face DistilBERT sentiment model against a local Ollama LLM (llama3.2:1b) returning structured JSON, with a client that routes between local and cloud inference
- Practiced prompt design for reliable structured LLM output and compared behavior across both approaches

Time-Series Forecasting with RNN/LSTM | TensorFlow/Keras | Time Series | 2026

- Forecast monthly milk production with SimpleRNN and LSTM models, using sequence windowing, scaling, and train/test splits that respect temporal order

Fashion-MNIST: MLP vs CNN | TensorFlow/Keras | Deep Learning | 2026

- Compared a flattened MLP baseline against a LeNet-5 inspired CNN; reported training curves, classification report, confusion matrix, and a model comparison table

NYC Airbnb Analysis & Machine Learning | Python | Scikit-learn | September 2026

- Cleaned and explored the NYC Airbnb listings dataset, trained and evaluated machine learning models, and presented methodology and findings

Video Game Sales Analysis | Python | Power BI | August 2026
- Team project: regional analysis of genres, publishers, platforms, and sales with Python EDA, a Power BI data model, and an interactive dashboard for business decision making

NFL Big Data Bowl 2027 (in progress) | Python | Spatiotemporal Tracking Data | Kaggle | Team entry
- Working with multi-season player tracking data (positions over time) to link combine measurements to NFL performance; writeup-judged, deadline January 2027

Real Time Hand Tracking Application | Python | OpenCV | MediaPipe | February 2026
- Real-time multi-hand detection with finger-position identification and dynamic finger counting

Virtual Voice Assistant | Python | NLP | API Integration | April 2026
- End-to-end voice assistant integrating SpeechRecognition, offline text-to-speech (pyttsx3), and external APIs (Wikipedia, YouTube) with multi-intent command parsing

CERTIFICATIONS

- **Samsung Innovation Campus AI Track Certificate** | Samsung & Life Makers Foundation | 2026
- **AI Fundamentals with IBM SkillsBuild** | IBM & Cisco Networking Academy | May 2026
- **AUC-SHE CAN Sustainable Innovators Boot Camp** | The American University in Cairo | August 2026: problem-first innovation, Business Model Canvas, investor-style pitch of a sustainability-focused project
- **Teach for Non-Tech Program** | I Career / Digitria | September 2025

TECHNICAL SKILLS

- **Languages:** Python (Advanced), C++ (Beginner)
- **ML and AI:** Scikit-learn, TensorFlow, Keras, NumPy, Pandas, Matplotlib, model evaluation, hyperparameter tuning, time-series forecasting, recommender systems
- **LLMs and Agents:** Hugging Face Transformers, Ollama (local LLMs), RAG, prompt engineering, agentic workflows, Make, n8n
- **MLOps and Tools:** MLflow, Flask, Streamlit, Gradio, Git, GitHub, VS Code, Google Colab, Kaggle
- **Data and BI:** Power BI, Tableau, Exploratory Data Analysis, Data Cleaning, Looker Studio, Excel
- **Computer Vision:** OpenCV, MediaPipe, image processing, CNNs
- **Soft Skills:** Analytical Thinking, Problem Solving, Teamwork, Leadership, Time Management, Entrepreneurship

LANGUAGES

Arabic: Native | English: Professional Working Proficiency (B2)